

Experiment No. 07

Decision Trees, Pruning and Overfitting

Aim:

To implement a **Decision Tree Classifier**, study the effect of tree depth on model performance, identify overfitting, and apply **cost-complexity pruning** to obtain a suitable decision tree with a good balance between accuracy and model complexity.

Objective:

After completing this experiment, students will be able to:

1. To understand the working principle of a **Decision Tree Classifier**.
2. To understand how a decision tree recursively partitions the dataset.
3. To study **Gini impurity and Entropy** as splitting criteria.
4. To train an unrestricted/deep decision tree.
5. To train a shallow decision tree and compare its performance with the deep tree.
6. To investigate the effect of max_depth on training and testing accuracy.
7. To identify **overfitting** using the train-test accuracy gap.
8. To understand the **bias-variance trade-off** in decision trees.
9. To apply **cost-complexity pruning** using ccp_alpha.
10. To select a suitable tree depth/pruning configuration based on accuracy and interpretability.

Pseudo Algorithm :

Steps:

1. Start
2. Load the classification dataset.
3. Inspect the dataset and identify the feature and target columns.
4. Handle missing values and encode categorical features if required.
5. Divide the dataset into training and testing sets.
6. Predict whether an unrestricted decision tree is likely to overfit.
7. Train an unrestricted Decision Tree Classifier.
8. Train a shallow tree with max_depth=3.
9. Calculate training and testing accuracy for both models.
10. Compare the train-test accuracy gap.
11. Vary max_depth from 1 to 14.
12. Train a separate decision tree for each depth.
13. Calculate training and testing accuracy at each depth.
14. Plot training and testing accuracy against tree depth.
15. Identify the depth at which the train-test gap begins to increase.
16. Generate the cost-complexity pruning path.
17. Obtain candidate ccp_alpha values.
18. Train a pruned decision tree using a suitable ccp_alpha.
19. Compare the pruned tree with the unrestricted tree.
20. Select the final depth/pruning configuration based on generalisation and interpretability.
21. Record observations and conclusion.
22. Stop

Sample Outcome:

- The decision tree classifier was successfully implemented using the Titanic dataset.
- An unrestricted decision tree and a shallow decision tree were trained and evaluated. The unrestricted tree achieved very high training accuracy but showed a larger difference between training and testing accuracy, indicating possible overfitting.
- The tree depth was then varied systematically. Training accuracy generally increased with tree depth, while testing accuracy improved initially and then either stabilized or decreased at higher depths.
- Cost-complexity pruning was subsequently applied to reduce unnecessary branches. The pruned tree provided a simpler model and helped control overfitting.

Sample Observation:

Deep vs Shallow Tree			
Model	Train Accuracy	Test Accuracy	Observation
Unrestricted Tree	1.00	0.72	High variance / possible overfitting
Shallow Tree (<code>max_depth=3</code>)	0.80	0.77	Better generalisation

Tools / Software Required:

Hardware: Computer/Laptop, Minimum 4 GB RAM

Software: Python 3.x, Jupyter Notebook or Google Colab, Anaconda (optional)

Python Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, Joblib / Pickle

Viva Questions:

1. What is a Decision Tree?
2. What is the difference between a classification tree and a regression tree?
3. What is Gini impurity?
4. What is entropy in a decision tree?
5. Why does an unrestricted decision tree tend to overfit?
6. What does max_depth control?
7. What happens if max_depth is too small?
8. What happens if max_depth is too large?
9. What is pruning?
10. What is ccp_alpha?